

CareerPilot AI

Precision Career Intelligence

From career confusion to a clear, evidence-based path toward employment.

A career readiness workspace for students, graduates, and early-career professionals.



Supporting documentation for project submission • September 2026

Live demo: <https://careerpiolet-ai.vercel.app>

Source code: <https://github.com/aminaahsan925/careerpiolet-ai>

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HOW TO READ THIS DOCUMENT

This document accompanies the live prototype and the public repository. Section 10, Winning Strategy, is written for hackathon judges: it states what makes this entry defensible under evaluation criteria and where the project goes next.

1. Executive Summary

CareerPilot AI helps students understand why they are unemployed and what they should do to become job-ready. Instead of offering generic courses or motivational advice, the platform compares a student's current profile with the expectations of a selected role and company. It turns identified gaps into a practical roadmap with daily learning, portfolio work, assessments, interview preparation, and recruiter-style feedback.

The project focuses on the gap between education and employability. Students may have degrees, certificates, and theoretical knowledge, but employers look for evidence: real projects, problem-solving, deployment, communication, and the ability to explain technical decisions. CareerPilot connects market reality, personal diagnosis, execution, and proof in one accessible workspace.

CORE PROMISE

Where am I now? Why am I being rejected? What does my target company expect? What should I learn, build, and practice next?

The prototype is deployed and end-to-end usable: a judge can register, complete onboarding, run a diagnosis, inspect market reality for a target role and company, generate a personalized roadmap, complete daily work, practice interviews, and receive recruiter-style feedback in a single session.

2. Problem and Audience

Students in Pakistan and other emerging markets often learn without a clear target. They do not know which skills matter for a specific job, why applications fail, what projects employers value, or how to prepare for interviews. Students outside major cities may also lack access to mentors, professional networks, and expensive career programs.

The result is a costly mismatch: months spent learning skills the market no longer asks for, resumes that never pass automated screening, and interviews that end at the first technical question. Existing tools address single fragments of this problem — job boards list openings, course platforms sell content, resume builders format text — but none of them connect a student's actual evidence to what a specific employer will actually screen for.

WHO IT IS FOR

- University students choosing a direction before graduation
- Recent graduates facing rejection without feedback
- Self-taught developers proving skills without a CS degree
- Career switchers entering tech from other fields
- Universities and training organizations seeking a structured employability journey

For institutions, CareerPilot offers a repeatable framework: every learner's readiness, gaps, and progress become measurable instead of anecdotal. This makes the platform relevant beyond individual use and into employability programs at scale.

3. Solution and User Journey

CareerPilot converts uncertainty into a sequence of decisions and actions:

01 Understand

Onboarding captures education, experience, skills, projects, certifications, and career goals.

02 Diagnose

The platform identifies missing skills, weak proof, outdated knowledge, and rejection risks.

03 Compare

Market Reality connects the student with role, company, Pakistan, and global hiring expectations.

04 Execute

A personalized roadmap provides daily learning, practical tasks, and portfolio-focused projects.

05 Practice

MCQs, interview preparation, and recruiter simulation turn knowledge into confident evidence.

06 Progress

Assessments, notifications, readiness, and completion tracking make progress visible.

Every stage feeds the next. The diagnosis defines the gap, the market reality defines the target, the roadmap defines the work, and assessments plus recruiter feedback verify that the work produced believable proof. No stage treats the student as a blank slate.

4. Key Capabilities

- Personalized career diagnosis with role and company context.
- Pakistan-focused Market Reality covering employer expectations, salary benchmarks, and practical opportunities.
- Day-by-day roadmap with learning explanations, hands-on tasks, and project evidence.
- Scenario-based MCQ assessments with scores, explanations, and attempt history.
- Interview preparation for technical, behavioral, project-defense, and company-specific questions.
- Resume Intelligence with ATS signals, structure review, detected skills, and career-fit analysis.
- Flight Plan readiness assessment that compares a live job description against recorded evidence.
- Recruiter Audit that explains how a student's profile may be screened, with persistent recruiter chat.
- Progress tracking, notifications, roadmap completion, and certificate unlocking.
- Responsive experience for desktop and mobile users.

Each capability operates on the same underlying career state, so evidence captured once — a project, a skill, a resume — strengthens every later analysis. This shared-state design is what keeps the advice specific instead of generic.

5. Innovation

CareerPilot is not only a course platform, chatbot, resume tool, or job board. Its innovation is the connection between diagnosis, market reality, learning execution, assessment, interview preparation, recruiter feedback, and proof. The platform helps a student move from knowing about a skill to demonstrating that skill in a way an employer can understand.

Three design decisions differentiate it from the fragmented tools students otherwise stitch together:

Evidence over claims

The system separates claimed skills from demonstrated proof. A student can list React, but the Recruiter Audit asks what deployed, measurable work backs the claim — the same question a hiring manager asks.

Local market grounding

Market Reality reports cite Pakistani on-site, remote-international, and global salary bands and expectations per role, so guidance is not imported wholesale from foreign job boards.

Research with citations

Live web research (Tavily) is cached per role and company. Findings carry sources, and claims fall back to the listed aggregators instead of invented numbers.

The result is a closed feedback loop: understand, diagnose, compare, execute, practice, and prove — with the market, not a static curriculum, as the reference point.

6. Technology and Architecture

The working prototype uses a modern full-stack architecture designed for maintainability, security, and future scale. The client uses React 19, TypeScript, TanStack Router, TanStack Query, Tailwind CSS, and Motion. Supabase provides authentication, Postgres persistence, storage, Row Level Security, user profiles, career goals, skills, projects, resumes, roadmaps, assessments, notifications, and recruiter sessions.

Server functions protect sensitive operations and keep AI credentials away from the browser. The AI layer supports Groq, Gemini, and OpenRouter providers with fallback handling, so a provider outage degrades gracefully instead of failing the user journey. Market and technology reports are cached to reduce repeated requests and improve loading performance. Server-side validation protects assessment results and user-scoped data.

LAYER	IMPLEMENTED APPROACH
Experience	Responsive React screens, mobile navigation, dashboard, roadmap, diagnosis, interview, recruiter, resume, and market workflows.
Application	TanStack Router, TanStack Query, reusable components, server functions, loading states, and cached queries.
Data	Supabase authentication, PostgreSQL persistence, user-scoped records, roadmap progress, MCQ attempts, and notifications.
Intelligence	Multi-provider AI generation for diagnosis, roadmaps, market analysis, recruiter feedback, and interview practice, with Tavily-backed live research.

Deployment is serverless on Vercel. All privileged keys are server-side environment secrets; only publishable Supabase credentials reach the browser. Every table and storage bucket is scoped to its owning user through RLS policies.

7. What Has Been Built

CareerPilot is a working, deployed prototype rather than a concept-only proposal. The implemented product includes authentication, onboarding, target role selection, Pakistani and international company selection, custom company entry, career diagnosis, market analysis, personalized roadmaps, daily tasks, MCQ assessments, interview preparation, recruiter simulation, resume analysis, notifications, progress tracking, certificate unlocking, and mobile-responsive layouts.

The surface area is intentionally deep rather than wide: thirteen authenticated routes cover the full journey, and each route operates on real persisted data rather than mock screens.

Diagnosis & Profile

Structured intake of education, experience, projects, skills, and blockers with market benchmark comparison.

Resume Intelligence

Upload with parsing, ATS signals, structure, strengths, weaknesses, detected skills, and career-match.

Flight Plan

Job-description readiness assessment combining role expectations, skill coverage, resume evidence, and market context.

Job Mirror

Employer expectations per researched role, separating claimed skills from demonstrated proof.

Recruiter Audit

Company-specific screening perspective with persisted sessions and recruiter conversation.

Roadmap

Personalized day-by-day learning with practical tasks, completion tracking, and certificates.

Market & Future Tech

Role salary benchmarks, demand signals, emerging tech radar, and outdated-tech warnings.

Mentor

Context-aware guidance initialized from the same career state as every other module.

8. Impact and Feasibility

CareerPilot can reduce wasted learning time, improve student confidence, and make career guidance more accessible. For universities and training organizations, it provides a structured employability journey. The architecture is modular, cloud-ready, and capable of expanding with more roles, employers, market sources, learning paths, and assessment content.

The near-term impact model is concrete:

- Time-to-readiness becomes visible: each student sees the distance between current proof and target-role expectations instead of guessing.
- Rejection becomes information: the diagnosis and recruiter audit convert 'we moved on with other candidates' into specific, fixable causes.
- Learning becomes accountable: daily roadmap work with completion tracking replaces open-ended course enrollments.
- Guidance becomes scalable: one mentor session is amplified by an AI layer that remembers each student's full context.

Feasibility rests on proven, low-cost infrastructure. Supabase's free tier covers hackathon-scale usage; AI inference is provider-agnostic with fallback; research results are cached per role and company to control API spend. The marginal cost of each additional student is low, and every AI response is designed to degrade gracefully when a provider is unavailable, keeping the core journey functional under constraint.

9. Demonstration Flow

A judge can understand the complete value of CareerPilot through this short product journey:

- 1 Create or open a student profile and complete onboarding.
- 2 Choose a target role and company, including a custom company.
- 3 Run a career diagnosis and review the rejection risks.
- 4 Open Market Reality to see what the selected role requires, with salary bands and sources.
- 5 Generate the personalized roadmap.
- 6 Open a roadmap day and complete the practical task.
- 7 Take the role-specific MCQ and review correct answers and explanations.
- 8 Practice an interview question and ask the recruiter simulator for feedback.
- 9 Review progress, notifications, and certificate completion status.

The full journey runs in under fifteen minutes on the deployed URL and requires no special preparation. A fallback demo account is available on request if registration emails are delayed.

10. Winning Strategy

This section states plainly why CareerPilot AI is a strong hackathon entry and how it is positioned to win. It is organized around the criteria judges typically apply: problem significance, working execution, technical depth, innovation, and presentation.

THE ONE-LINE PITCH

CareerPilot AI is the only entry that closes the loop from evidence to employability: diagnose the gap, ground it in the live market, execute daily, and prove it to a recruiter — all shipped and deployed.

Five strategic advantages

1. A real problem, precisely framed

Graduate unemployment in Pakistan is not abstract; it is measurable and personal. CareerPilot addresses it with a specific mechanism — evidence-based readiness — rather than another content library. Judges can verify the problem with one question to any student.

2. Shipped, not slideware

The product is deployed end-to-end with authentication, persisted data, RLS-secured tables, and thirteen working routes. Every claim in this document is clickable on the live URL. In a field of mockups, a working system is the fastest trust signal.

3. Technical depth under a simple UX

Beneath a clean interface sits real engineering: multi-provider AI fallback, cached Tavily research with citations, server-only credential handling, server-validated assessments, and a schema with per-user row-level security. Depth is documented in Section 6 and visible in the repository.

4. Local grounding with global standards

Market Reality speaks to Pakistani salary bands and hiring realities while tracking global frontier tech. This dual lens matches the hackathon theme — AI built for Pakistan's future — without lowering the technical bar.

5. A demo that tells a story

The nine-step demonstration flow (Section 9) walks a judge from confusion to certificate in one sitting. Every screen answers the previous screen's question, which keeps attention and makes the value self-evident.

Risk mitigation and honesty

No fabricated numbers

Market figures come from cited research with named sources; when live data is unavailable, the system falls back to the listed aggregators rather than inventing values. Credibility is treated as a feature.

Graceful degradation

If an AI provider rate-limits during judging, the platform degrades to cached research and fallback analysis instead of erroring out. The demo cannot be killed by a third-party outage.

Scoped ambition

The prototype deliberately executes one journey deeply instead of many shallow ones. Post-hackathon expansion (Section 11) is sequenced, not hand-waved.

11. Why CareerPilot Matters

CareerPilot responds to a practical national challenge: helping educated young people become employable through better information, focused learning, and credible evidence. It supports students who do not know where to begin and gives them a way to understand their profile from an employer's perspective.

CLOSING STATEMENT

CareerPilot AI helps every student understand the gap between their current profile and the job they want, then gives them a practical path to close that gap with real evidence.

Post-hackathon roadmap in three phases

Phase 1 — Depth

Expand role coverage and company research, deepen interview question banks, and open recruiter audit to more markets.

Phase 2 — Reach

University and bootcamp partnerships for cohort-based employability tracking, with mentor dashboards for institutions.

Phase 3 — Proof at scale

Longitudinal outcome tracking: correlate roadmap completion with interview pass rates and offers, turning CareerPilot's promise into measured results.

LINKS

Live demo	https://careerpiolet-ai.vercel.app
Source code	https://github.com/aminaahsan925/careerpiolet-ai

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